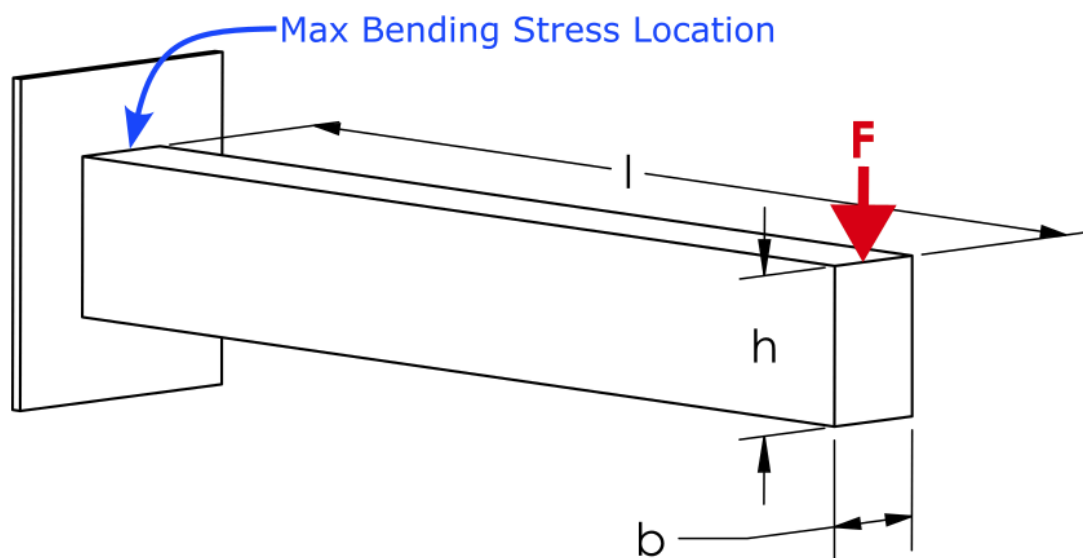


Database Consistency Test

Introduction

This EngineeringPaper.xyz sheet shows how to calculate the maximum stress and displacement for a cantilever beam loaded by a vertical force at its end. The geometry of the beam is shown in the figure below. The beam has a rectangular cross section with height h and width b . The maximum stress in a cantilever with length much larger than its height is due to bending stress that occurs at the base of the beam. For a downward force, the maximum stress will be a tensile force at the top of the beam and a compressive stress at the bottom of the beam. For beam sections that are symmetric top to bottom, the maximum tensile and compressive stresses will have equal magnitude.

μ ρ 你好吗？请问你想把“cats and dogs”翻译成普通话吗？ 🤖😄😊🌴🌳🌲 #



Select Beam Material

$$\text{Aluminum Alloys} \quad \begin{cases} E = 71.7 [GPa] \\ v = 0.34 \\ \rho = 2800 \left[\frac{kg}{m^3} \right] \end{cases}$$

Source: Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

Select Beam Section

$$\text{Rectangle} \left\{ \begin{array}{l} I_x = \frac{b \cdot h^3}{12} \\ c = \frac{h}{2} \\ r = [mm] \\ r_1 = [mm] \\ r_2 = [mm] \\ a = [mm] \\ b = 50[mm] \\ h = 75[mm] \\ b_1 = [mm] \\ h_1 = [mm] \end{array} \right.$$

Source for beam section equations and images: https://en.wikipedia.org/wiki/List_of_second_moments_of_area

Define the Other Beam Parameters

$$l = 500[mm]$$

$$F = 500[N]$$

Define the Deflection and Stress Equations

The maximum displacement of a cantilever beam loaded at its end is defined by [1]:

$$y_{max} = -\frac{F \cdot l^3}{3 \cdot E \cdot I_x}$$

where I_x is the area moment of inertia of the cross section about the x-axis as selected in the table above. The maximum displacement can now be queried:

$$y_{max} = [mm] = -0.165297794307557[mm]$$

The maximum bending stress in a beam section is given by:

$$\sigma_{max} = \frac{M \cdot c}{I_x}$$

where M is the bending moment at the section of interest and c is distance from the neutral axis of the beam to the top or bottom of the beam. The maximum moment occurs at the base of the beam and is defined by:

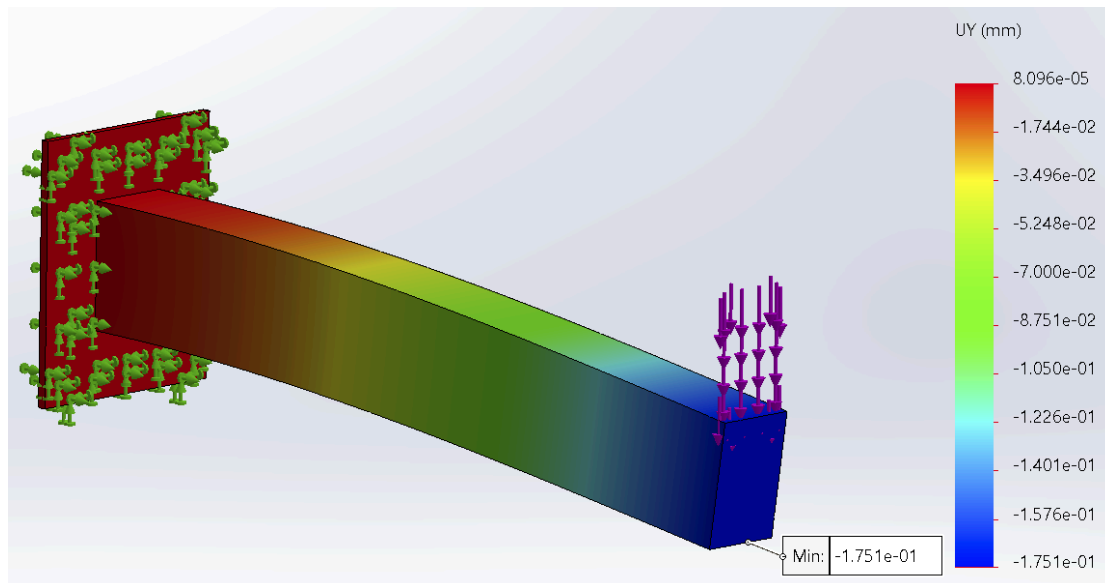
$$M = l \cdot F$$

where l is the length of the beam. Now that everything is defined, the maximum stress can be queried and converted to [MPa] units:

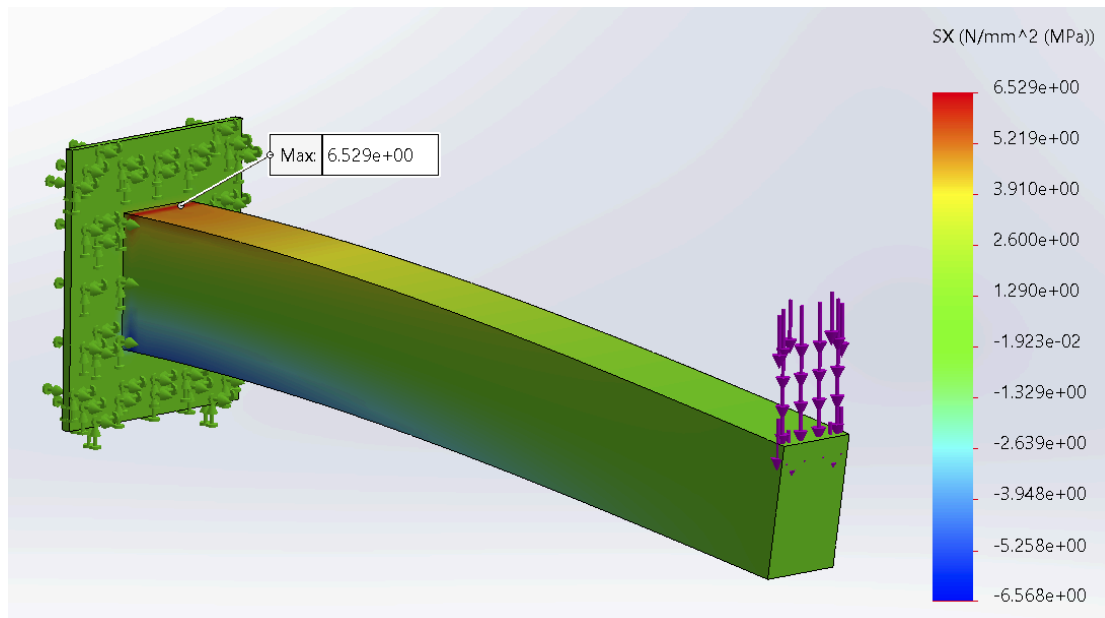
$$\sigma_{max} = [MPa] = 5.33333333333333[MPa]$$

These results can be compared to the results obtained from a finite element analysis using SolidWorks. The beam has the same dimensions, loading, and material as the rectangular cross-section and the aluminum table options above. The displacement plot

is shown below. The maximum displacement magnitude obtained using finite element analysis is .175 [mm], which is close to the .165 [mm] obtained using the equations above.

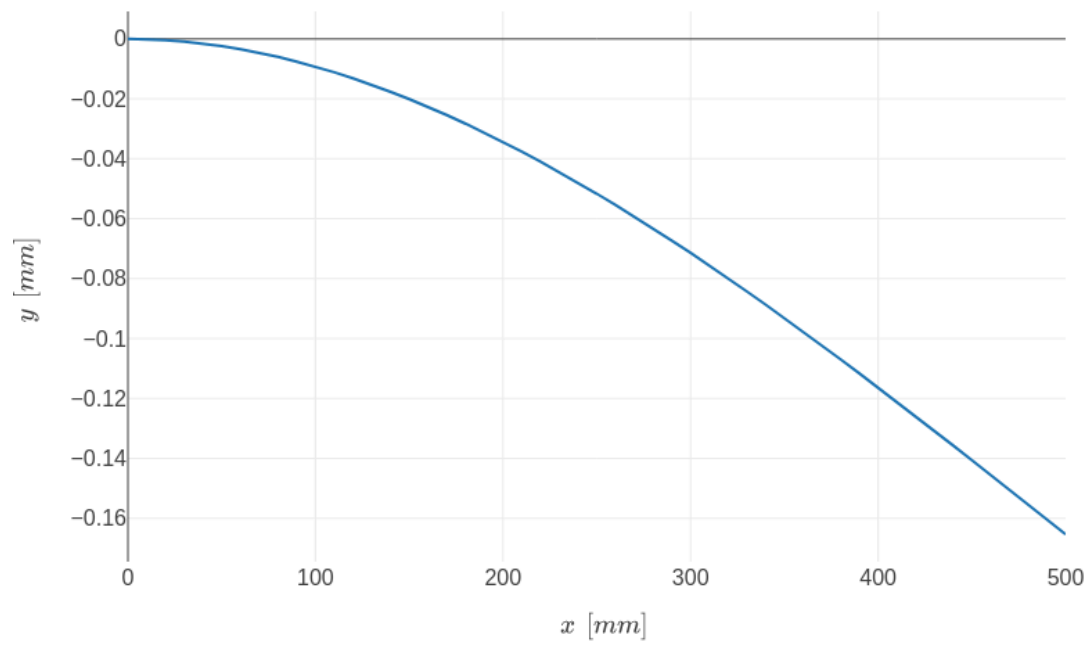


Similarly, the maximum stress calculation can be compared to the SolidWorks results. The maximum stress obtained from SolidWorks is 6.5 [MPa] as shown below. Note that this is higher than the 5.3 [MPa] obtained using the beam equation. This is the result of the stress concentration that occurs at the base of the cantilever where it meets the fixed boundary condition. A little further from the fixed boundary condition, the finite element values are close to the calculated values.



[1] Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

$$y = \frac{F}{6 \cdot E \cdot I_x} \cdot (x^3 - 3 \cdot l \cdot x^2)$$



All Colors

one

two

three

four

five

six

seven

nine

ten

eleven

twelve

thirteen

fourteen

fifteen

sixteen

seventeen

eighteen

nineteen

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MD Export Test (centered equations unchecked)

$$x = 20[m]$$

Documentation cell with line of text followed by a scaled image:



Sentence with formula $\sqrt{\Omega}$ in the middle, followed by equation on its own line:

$E = \frac{My}{I}$

With text after.

Line of text.

Another line of text.



Option 2 $\begin{cases} l_1 = 3[mm] \\ l_2 = 4 \end{cases}$

$l_2 = 4$

$l_1 = [m] = 0.003[m]$

<i>Col1</i>	<i>Col2</i>
	[m]
6	7
8	9

$y = \begin{cases} 10: & x > 20 \\ 2 : & \text{otherwise} \end{cases}$

System = $\begin{cases} 1 \cdot s + 2 \cdot t = 10 \\ 2 \cdot s - 4 \cdot t^2 = 20 \end{cases}$

Solution = $\begin{cases} s = 12 \\ t = -1 \end{cases}$

Roller Thrust Bearing $\mu_c = 0.01$

This left justified, below is a centered equation:

$x = 2 \cdot y$

Below is a centered image:



This is right justified

This is justified text

- red
 - blue
1. one
 2. two
 - a. three
 - b. four

Normal text

MD Export Test (centered equations checked)

$$x = 20[m]$$

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Sentence with formula $\sqrt{\Omega}$ in the middle, followed by equation on its own line:

$$E = \frac{My}{I}$$

With text after.

Line of text.

Another line of text.



$$\text{Option 2} \quad \begin{cases} l_1 = 3[mm] \\ l_2 = 4 \end{cases}$$

$$l_2 = 4$$

$$l_1 = [m] = 0.003[m]$$

Col1	Col2
	[m]
6	7
8	9

$$y = \begin{cases} 10 : & x > 20 \\ 2 : & \text{otherwise} \end{cases}$$

$$\text{System} = \begin{cases} 1 \cdot s + 2 \cdot t = 10 \\ 2 \cdot s - 4 \cdot t^2 = 20 \end{cases}$$

$$\text{Solution} = \begin{cases} s = 12 \\ t = -1 \end{cases}$$

$$\text{Roller Thrust Bearing} \quad \mu_c = 0.01$$

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Normal text

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